PEANO ARITHMETIC / TURING MACHINES

PA194QN

Exchanging premises without a dummy stack push

A distinct 194-state binary consistency machine,
with a complete transition table and reproducible evidence

Result. No verified construction below 194 states was obtained. The delivered machine has 194 working states and 388 transitions and is absent from the 34 existing 194-state graph classes in the pinned corpus. Its new preservation argument composes the Q axiom emitter with the N no-push controller. Using the explicitly inherited numerical-calculus/PA calibration, the ordinary mathematical conclusion is

$$\boxed{\mathrm{PA} \vdash (\neg\,\mathrm{Halt}(\mathsf{PA194QN}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})).}$$

Verification level. The archive includes universal symbolic physical checks, an independent source and logical audit, and complete blank-tape initialization. It does not supply a newly compiled end-to-end Lean/Rocq theorem or an internal iPRA proof of this equivalence.

Repository snapshot

`3cc7fbf75ba111a0da696ef2f6001dd528891bd5`

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The sources, provenance, comparison ledger, verification code, receipts, and optional unsuccessful searches accompany this report.

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1 Outcome and exact machine convention

The primary target was a machine with fewer than 194 working states. The completed searches did not establish one. The fallback result is the fully specified machine PA194QN. Its state count is

$$134 \text{ fixed controls} + 67 \text{ axiom-reader controls} - 7 \text{ simultaneous folds} = 194.$$

The tape alphabet is $\{0, 1\}$, the tape is two-way infinite, and every cell is initially zero. The head begins at coordinate zero in state zero. Working states are $0, \ldots, 193$; label 194 is terminal and has no outgoing rows. Every defined transition writes one bit and moves exactly one cell, left or right. All 194 working controls are reachable in the transition graph. Graph reachability is not a claim that every control is visited on the actual blank computation.

The authoritative file is `machine/machine.tm`. Its SHA-256 is

`d1cf0f6e7a3c8f999d045694efc107dd1f9a03d45dd6c15e3d197ff7754c7231`

Appendix A prints the entire table. A row in the machine file has the form `state read write direction next`. Native labels, the dense-label bijection, the unfolded 201-control parent, and every source instruction are also supplied.

1.1 What is new, and what is inherited

The Q emitter[3] exchanges the two curried premises of the universal distribution head. The N controller performs successful modus ponens without pushing a dummy zero, retaining the result in the visible top and in the offset[2]. PA194QN combines these two mechanisms while retaining N's original Shift guard $t + h = 1$. Neither constituent idea is claimed as new in isolation. The new object is their exact 194-state realization, its compatible source and command interface, its preservation proof, and its verified table.

The combination is substantively different from both parents. Relative to PA194N, it changes an emitted formula and 79 source instructions. Relative to PA194Q, it changes both the MP result convention and the Shift guard. For example, $I(0, 2) = 20$, and MP with parameter two sends

$$(0, w, 0, 20) \longmapsto (2, w, 2, 0)$$

in PA194QN, whereas Q's cached controller sends that store to $(0, w, 0, 2)$. Also, PA194QN's Shift halts from $(1, 0, 0, 0)$, whereas the cached-only guard does not. These are differences of command behavior, in addition to the literal graph distinction.

The finite novelty check compares complete transition tuples, allowing arbitrary state renaming, any graph-spanning root choice, left/right reflection, and symbol complementation. It finds no match among all 34 recognized existing 194-state graph classes. This statement is explicitly scoped to the finite literal-table corpus; it does not assert that the combination was absent from every possible informal idea or source-only encoding. In particular, a prior NS search already screened Q and D emitters under the different NS controller.

2 Comparison of the supplied designs

The comparison covers the PA research tree and all four current incoming ZIPs at the pinned snapshot[1]. The narrative ledger has 51 entries: 47 placed research packets and four incoming deliveries. The literal recognizer accepts complete numeric five-column binary tables in `.tm`, `.txt`, `.tsv`, and `.csv` files and traverses nested ZIPs.

Fresh inventory quantity	Count
Recognized table occurrences	1,811
Distinct table bytes	683
Whole-table action graphs with a spanning root	487
Distinct ZIP payloads / archive references	18 / 30
194-state occurrences / byte forms / graph classes	208 / 62 / 34
195-state graph classes	17
196-state graph classes	99
Scan errors	0

These counts describe data, not successful consistency constructions. Among the 34 classes at 194 states, the supplied evidence supports 21 ordinary mathematical PA arguments with physical evidence; nine are auxiliary tables and four are explicitly rejected experiments. This classification does not constitute a fresh independent proof of every historical claim. The two recognized smaller synthetic byte forms are test controls, not consistency-machine results.

The repository's existing census reports 481 classes under a different scope. Its classes reconcile to 478 whole-table spanning-root graphs and three reachable-subgraph classes of disconnected synthetic controls. The current incoming material adds nine whole-table graphs: four at 194, one experimental 195-state table, and four larger unfolded parents. The supplied reconciliation file explains every old class. The original 3,803-member delivered-table count is a historical count before duplicate removal; it should not be compared directly to 1,811 currently recognized occurrences.

2.1 Design families and the sources of savings

Family	Comparison and significance
U208 / PA209	Entry context, finite history, and local decoder folds produce several presentations, with substantial duplication between packets. These methods are ancestors of the later contextual quotients.
PA196 variants	Reassociation, diagonal vacuity, a separate diagonal head, cached results, known-zero jumps, and scanner changes alter different layers. A shorter register program need not yield a smaller physical TM.
Original PA195	The bundled-diagonal emitter has 191 macro tokens; its 68 readers and seven folds give $134 + 68 - 7 = 195$.
PA195S / C / DC	Separate-head PA195S has 190 macro tokens and $134 + 67 - 6 = 195$. C changes the result controller; DC combines the bundled emitter with the cached controller.
M195S8	An extra scanner control repairs a differently associated source. The associated nominal 194-state table with the shorter scanner is rejected by a source-frame counterexample.
PA194 / AD / 68–8	The main 67-reader/seven-fold allocation, its physical A/D register permutation, and a distinct 68-reader/eight-fold allocation are different tables with the same principal calculus.
PA194C / CS	The result is cached in h, Shift tests $h = 1$, and the proof compiler changes accordingly. CS makes the comparison symmetric and uses an additional controller fold.

Family	Comparison and significance
M4R / M5 / M6	M4R changes association. M5/M6 change the constructor coefficient and require explicit formula and proof recoding. They do not obtain their meaning from merely changing a multiplier in the table.
PA194D / Q	D changes the diagonal head to $\Gamma(D)$ and recompiles its selector. Q exchanges premises in universal distribution and supplies an implicational proof translation.
J / A / F / CSZ / ZR	Selected source jumps become decrements of registers proved zero at those points. The zero facts must hold on the actual source execution and remain compatible with physical startup.
PA194R	A new association is combined with a final known-zero decrement; the source paths change while the resolved instruction graph is retained.
PA194N	Successful MP omits its dummy-zero push. The new proof compiler reserves and consumes the needed zero explicitly; Shift remains $t + h = 1$.
H / HS / NS	H and HS combine visible-top no-push MP with a cache-only Shift guard. HS uses symmetric comparison. HS and NS are the same dense table despite their different delivery names.
N5 / N6	Coefficient-five/six constructor recoding is combined with no-push MP. The two N6 incoming submissions contain identical dense bytes.

The full 34-class frontier appears in Appendix B; the 51-delivery ledger appears in Appendix C. Machine-readable versions include exact locators, hashes, and representative tables. Five packets repeat the principal PA194C graph; PA194X repeats PA194CSZ. These duplicates are retained in the delivery ledger but not counted as new designs.

2.2 Proof levels in the repository

The current PA README explicitly separates ordinary mathematical PA calibrations from the unfinished machine-specific internal iPRA reductions. The imported PA196 Lean endpoint includes the implication from literal nonhalting to PA consistency; that fact alone is not the converse or an iPRA equivalence. A finite-run quotient theorem, a Python certificate, and a conditional arithmetic endpoint each address different proof obligations. This report retains that separation throughout.

3 Numerical formulas and the changed axiom

Use Cantor pairing and constructors

$$p(a,b) = (a+b)(a+b+1)/2 + b, \qquad I(A,B) = 4p(A,B),$$
$$M(a,b) = 4p(a,b) + 1, \quad U(a,B) = 4p(B,a) + 2, \quad E(a,b) = 4p(a,b) + 3.$$

Code zero denotes truth; code one is the contradiction $M(0,0)$. Write $N(B) = I(B,1)$, $B \wedge C = N(I(B,N(C)))$, and $\exists_a B = N(U(a,N(B)))$. The zero alias $I(0,0) = 0$ is real and must be handled as an equality of numerical constructors, not denied by an assumption that formulas are a free term algebra.

For parameters D, A, H, the six ordinary emitted heads, in emission order, are

$$\begin{aligned}
L_1 &= I(\exists_D U(D,H),H),\\
L_2 &= I(U(D,U(A,H)),U(A,U(D,H))),\\
L_3 &= I(M(H,D),I(E(H,A),M(A,D))),\\
L_4 &= I(E(H,D),I(M(A,H),M(A,D))),\\
L_5 &= \exists_D E(H,D),\\
L_6' &= I(U(D,I(A,H)),I(U(D,A),U(D,H))).
\end{aligned}$$

The old sixth head was

$$L_6 = I(U(D,A),I(U(D,I(A,H)),U(D,H))).$$

The additional heads are the diagonal instance $\Gamma(A) = I(A,U(A,A))$ and the dominated implication instance

$$\rho(D,A,H,d) = I(I(I(H,d),D),I(I(D,H),I(A,H))), \qquad d \le D.$$

The guard and separation gate are unchanged. In fully explicit notation,

$$\begin{aligned}
V &= I(M(H,D),U(A,M(H,D))), & P &= I(A,M(A,D)),\\
B &= U(D,I(M(H,D),P) \wedge I(M(A,D),A)), & Q_s &= U(A,I(B,N(P))),
\end{aligned}$$

$$g(H,D,A) = \begin{cases} N(I(V,Q_s)), & A \neq H \text{ and } A \neq D,\\ 0, & \text{otherwise.}\end{cases}$$

Lemma 3.1 (Primitive-recursive logical transport)**.** *Replacing the universally instantiated head L_6 by L_6' leaves the numerical theorem set unchanged. There are conclusion-preserving primitive-recursive translations of finite derivations in both directions, and PA proves their correctness.*

Proof. The unchanged dominated implication basis derives K, S, and premise exchange

$$I(I(P,I(Q,R)),I(Q,I(P,R))).$$

One obtains the reverse implication by interchanging P and Q. Substitute $P = U(D,A)$, $Q = U(D,I(A,H))$, and $R = U(D,H)$, then apply MP. At a changed axiom occurrence, replace that leaf by this fixed proof template; preserve all other proof constructors. Structural recursion and induction on the coded proof give the stated translations. The derivation of exchange uses only the dominated implication basis and does not use either version of universal distribution. It is therefore not circular.

The archive includes a separately replayed 61-line exchange certificate: 17 K instances, 14 S instances, and 30 MP steps, with no assumptions. The inherited derivations of K and S are identified in `references/ImplicationBasis.lean` and `references/RestrictedImplication.lean`. Because the proof template uses only numerical implication formation and MP, it remains valid in the case $I(0,0) = 0$. □

The formula change is numerically visible. For $D = A = 0, H = 1$ the three subformulas (P, Q, R) have codes $(2, 146, 6)$, and L_6 and L_6' have different codes. The change is not a global permutation of formula variables or a relabeling of the TM's controls.

4 The no-push command interface and its proof compiler

Let $[a; w] = p(a, w)$, with total projections hd and tl. Thus $[0; 0] = 0$ and both projections of zero are zero. Iterated brackets have their leftmost item at the top. A store is (t, w, d, h): visible top, proof stack, numerical offset, and cached head.

For $b = d + x$, the MP and Shift commands are

$$\mathsf{MP}_x(t, w, d, h) = \begin{cases} (b, w, b, 0), & I(t, b) = h, \\ (0, w, b, 0), & I(t, b) \neq h, \end{cases}$$

$$\mathsf{S}(t, w, d, h) = \begin{cases} \mathsf{halt}, & t + h = 1, \\ (h, \operatorname{tl} w, d, \operatorname{hd} w), & t + h \neq 1. \end{cases}$$

MP success preserves w exactly. No cell is implicitly pushed by this operation. For $D = d + x$, $H = h + y$, AX returns

$$\mathsf{AX}_{A,x,y}(t, w, d, h) = \big(g(H, D, A), [q_1, \ldots, q_9, t; w], 0, 0\big),$$

where

$$(q_1, \ldots, q_9) = (\Gamma(A), L_6', L_5, L_4, L_3, L_2, L_1, \rho(D, A, H, d), U(d, h)).$$

All of $g, q_1, \ldots, q_8$ have residue zero modulo four, and $q_9 = U(d, h) \geq 2$ has residue two. None is one, for arbitrary input parameters, without assuming consistency. The free-algebra AX check verifies the entire stack and gate, field reads and cleanup; it does not merely check these residues.

4.1 Total stack identities

Words execute from left to right. Direct calculation gives

$$\mathsf{AX}_{A,x,y}; \mathsf{S}^9(t, w, d, h) = (q_8, [t; w], 0, U(d, h)),$$
$$\mathsf{AX}_{A,x,y}; \mathsf{S}^{10}(t, w, d, h) = (U(d, h), w, 0, t).$$

These Shifts are safe because their tests include generated non-one residues, or the universal code of value at least two.

Put $A_0 = \mathsf{AX}_{0,0,0}$ and define

$$\mathsf{Push} = A_0; \mathsf{S}^9; \mathsf{MP}_0, \quad \mathsf{Pop} = A_0; \mathsf{S}^{11}, \quad \mathsf{Discard} = \mathsf{MP}_0, \quad \mathsf{Offset}_i = \mathsf{MP}_i.$$

Their exact contracts are

$$\mathsf{Push}(t, w, d, h) = (0, [t; w], 0, 0),$$
$$\mathsf{Pop}(t, [a; w], d, h) = (t, w, 0, a),$$
$$\mathsf{Discard}(t, w, 0, h) = (0, w, 0, 0),$$
$$\mathsf{Offset}_i(0, w, 0, 0) = (0, w, i, 0).$$

For Push, the final MP compares an implication with a universal code and therefore fails with zero offset. Pop's last Shift tests $U(d, h) + t \geq 2$, so Pop is total even when $t = 1$. Discard works for arbitrary old cache: with $d = x = 0$, both outcomes of MP equal $(0, w, 0, 0)$. Offset works also at $i = 0$, where the same two outcomes coalesce.

These are numerical projection identities. They remain true when a zero push collapses into an empty numerical stack. In particular, $\mathsf{Pop}(t, 0, d, h) = (t, 0, 0, 0)$. No proof below depends on a nonexistent distinguishable zero stack cell.

4.2 Selectors, generalization, and MP

For an arbitrary incoming $(t, w, 0, 0)$ and $1 \le j \le 7$, the selector

$$\mathsf{Discard}; \mathsf{AX}_{A,D,H}; \mathsf{S}^{j+1}; \mathsf{Pop}^{9-j}$$

returns $(q_j, w, 0, 0)$. The initial Discard makes the saved old top zero. The Shifts select q_j; the Pops remove the unselected tail and saved zero. The gate uses $\mathsf{Discard}; \mathsf{AX}_{A,D,H}; \mathsf{Pop}^{10}$. For $u \le a$, the dominated-head selector is

$$\mathsf{Discard}; \mathsf{Offset}_u; \mathsf{AX}_{b,a-u,c}; \mathsf{S}^9; \mathsf{Pop}.$$

It returns $(\rho(a, b, c, u), w, 0, 0)$. Truth is supplied by Discard. The exchange translation in Section 3 supplies the old sixth head whenever a proof calls for it.

Generalization is compiled by

$$\mathsf{Gen}_i = \mathsf{Push}; \mathsf{Offset}_i; \mathsf{S}; \mathsf{AX}_{i,0,0}; \mathsf{S}^{10}.$$

It sends $(a, w, 0, 0)$ to $(U(i, a), w, 0, 0)$, including $a = 1$. Before its AX, the store is $(0, w, i, a)$; all the displayed Shifts are safe.

At an MP node with minor a, major $I(a, b)$, and recursive proof words $C(a)$ and $C(I(a, b))$, use

$$\mathsf{Discard}; \mathsf{Push}; C(I(a, b)); \mathsf{Push}; C(a); \mathsf{Push}; \mathsf{S}; \mathsf{S}; \mathsf{MP}_b; \mathsf{Pop}.$$

The first two commands reserve a zero below the premise computations. If no recursive execution has halted, the suffix begins at $(0, [a, I(a, b), 0; w], 0, 0)$. Two Shifts either halt on $a = 1$ or establish $(a, [0; w], 0, I(a, b))$. MP returns $(b, [0; w], b, 0)$, and the final total Pop returns $(b, w, 0, 0)$, including $b = 1$. The final Pop is essential: the single Shift appropriate to Q's cached-result compiler would not implement this interface.

Lemma 4.1 (Uniform proof compiler)**.** *For every finite derivation p in the changed calculus, the recursively compiled word $C(p)$, starting from any $(t, w, 0, 0)$, either halts early or returns* $(\mathrm{concl}(p), w, 0, 0)$.

Proof. Induct on the proof tree using the displayed selectors, generalization identity, and reserved-zero MP suffix. Each recursive word preserves the arbitrary underlying stack. Early halting is preserved by word extension. The calculations include the cases $I(0, 0) = 0$ and $[0; 0] = 0$. □

If p proves one, the word $A_0; C(p); \mathsf{S}$ halts from *every* incoming store. AX first clears the offset and cache. If no earlier halt occurs, the final Shift tests $1 + 0 = 1$. This all-store contract is needed because the scheduler carries its store between failed trials.

5 From finite proofs to a PA consistency equivalence

5.1 The exhaustive scheduler

The source scheduler is inherited unchanged. With positive continuation coding $E(\epsilon) = 0$ and $E(c :: w) = 1 + B_c(E(w))$, its command encodings are

$$B_{\mathsf{S}}(z) = p(0, z), \quad B_{\mathsf{MP}_x}(z) = p(1, p(x, z)), \quad B_{\mathsf{AX}_{A,x,y}}(z) = p(A + 2, p(y, p(x, z))).$$

Let $z = E(A_0; C(p); \mathsf{S})$. Prefixing this all-store halting word by $\mathsf{AX}_{k,0,0}$ gives fresh-candidate codes

$$n_k = p(k + 2, p(0, p(0, z))).$$

They are strictly increasing and unbounded. The fresh counter begins at two, so omission of zero and one loses none of this cofinal family. For a positive continuation l, the next decoded residual is at most $l - 1$. Each trial therefore finishes or halts; the inherited arithmetic macro loops have primitive-recursive terminating bounds. Finite induction through earlier candidates reaches a selected n_k, unless a halt has already occurred. The outer fresh-candidate code and the positive continuation code differ by the displayed convention; conflating them would invalidate this argument.

5.2 Soundness of the command execution

Annotate the visible top, cached head, and every live canonical stack entry with a numerical-calculus derivation. Code zero has a fixed truth derivation; it also annotates the virtual zero tail. For a nonzero Cantor stack, its tail is strictly smaller than the stack code, giving a bounded primitive-recursive construction of the canonical annotation. No theorem annotation is imposed on the offset d.

AX adds valid heads, a generalization of the annotated old cache, and the saved old top. On successful MP, the annotations of t and $I(t, b) = h$ give a derivation of b; failure returns truth. Shift transports the annotations. If its guard succeeds, $t + h = 1$ implies that one of the two annotated values is code one. Thus a source halt yields a finite refutation in the numerical calculus. The proof updates are primitive recursive and are verified by ordinary induction in PA.

5.3 The inherited numerical-calculus calibration

The supplied separate-head calibration[4] provides PA-verifiable proof-producing translations between PA refutations and derivations of numerical code one. Its mathematical texts are included in `references/logical/`, with their source identities. This is an explicit dependency of the result, not something inferred from finite simulation. The pertinent parts are:

1. Interpret $M(a, b)$ by finite binary membership

$$\mathrm{Bit}(s, z) \equiv \exists u, v\, [s = (2u + 1)2^z + v \ \wedge \ v < 2^z],$$

   and E by equality. PA proves $\mathrm{Bit}(s, z) \to z < s$, hence refutes the translation of code one.
2. For each arithmetic formula θ, induction produces a PA proof of finite binary comprehension

$$\forall n\, \exists s < 2^n\, \forall z < n\, (\mathrm{Bit}(s, z) \leftrightarrow \theta(z)).$$

   Substitution into this proof template is a syntactic primitive-recursive operation. It does not assert a uniform arithmetic truth predicate.
3. Every variable label in a decoded formula of code B is below B; the case $B = 0$ is handled separately. Therefore $\Gamma(B)$ has the required fresh binder. This supports the replacement of the bundled diagonal gate by its separate head.
4. The separation translation distinguishes the coincident-label case from distinct labels. When labels coincide, the binder changes both occurrences; the finite-comprehension matrix must change accordingly.
5. The completeness translation derives reverse membership rewriting and neighborhood equality with fresh diagonal binders, then uses the original refutation compiler. The removed equality-vacuity schema is not silently assumed.
6. $I(0, 0) = 0$ is reconciled by fixed proof templates relating truth and truth implying truth. It is not treated as literal free-syntax injectivity.

The new exchange translation preserves these calibrations. The all-store compiler and the scheduler then give, in PA, a finite source halt from every PA refutation. Conversely, the annotation construction and inherited arithmetic translation turn a source halt into a PA refutation. Thus

$$\mathrm{PA} \vdash (\mathrm{Halt}(R_{QN}) \leftrightarrow \exists p \, \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner)) \, .$$

No consistency assumption or reflection principle for PA is used to prove this biconditional. It is a statement about finite proof and computation codes. The actual nonhalting of the resulting machine is conditional on the consistency of PA.

6 Binding the source to the exact 194-state table

6.1 Source change and allocation

The inherited source has nine registers $(N, T, W, A, D, H, X, Y, L)$. The logical-to-physical register map is $(6, 2, 0, 4, 3, 7, 8, 5, 1)$. The Q grammar has 190 macro tokens; its changed block is the interval $[102, 121)$ in zero-based token positions. The whole resolved QN source has 3,654 sites. Relative to N, exactly 79 instruction rows change, all within indices 1,953 through 2,067 inclusive. All other instruction rows coincide.

The N splice replaces

`node163=(node6,node55)`, `node55=(node49,node54)`

by `node163=(node6,node54)`. It deletes a 19-site dummy-zero push interval $[3634, 3653)$ from the corresponding push-based Q source. The interval has one external entry, from old instruction 3628. The builder checks every surviving instruction and successor after exact index relocation. The native physical edit is

$$(164, 1) : (1, R, 98) \longmapsto (1, R, 114).$$

At successful MP, the retained copy loop begins with $D = b, T = X = 0$. Its first loop maintains $(T, D, X) = (j, b - j, j)$ for $j \leq b$; the second maintains $(T, D, X) = (b, j, b - j)$. It exits with $(b, b, 0)$ in $7b + 2$ primitive steps. This explains the exact result and offset convention used in the command proof.

The grammar compiles to 67 readers atop 134 fixed controls. Seven disjoint contextual pairs are folded. Each removed control is entered after a preserved zero to its left; its partner is entered after a preserved one. On the final simultaneous table, all four combinations of history bit and current read bit are checked per pair. The local detour reaches exactly the expected aliased state, head position, and tape. The archive contains all 28 traces and all incoming rows. Shared helpers are therefore checked after all folds have been applied, not only against the parent table.

6.2 Universal physical verification

Reproduced quantity	Value
Working controls / transition rows	194 / 388
Primitive source sites / outcomes	3,654 / 4,755
Simultaneous fold cases	28
Construction symbolic steps	1,506,202
Construction maximum per outcome	488
Independent symbolic steps	1,102,319
Independent maximum per outcome	397
Minimum physical-duration constant	3
Generic list lemmas rederived from literal rows	4
Exact blank-start transitions to clean frame	58,154

Each primitive outcome is checked for nine arbitrary natural register values, both decrement branches, an arbitrary natural affine gap, and an arbitrary nonempty finite list of arbitrary natural spare values. The physical header length is $48 + n$ with arbitrary $n \geq 0$. The construction checker uses the unchanged arithmetic-block interfaces. The separate independent checker rederives four generic list lemmas from the delivered rows and uses no arithmetic-box oracle. Two independently implemented source resolvers agree with the exported graph. A further audit expands saved source nodes and resolves them by subtree ordinal intervals without importing the builder or physical simulator.

Startup is checked with two concrete representations. The entire tape, state, and head agree after 58,154 transitions: native state 84, head -2, source site 25, logical registers $(2, 0, \ldots, 0)$, and 26 spare blocks of value one. Every nonzero cell is compared. A startup trajectory is only an initialization certificate; its finite length is not evidence for infinite nonhalting.

Lemma 6.1 (PA-verifiable finite simulation)**.** *The delivered table halts from its blank initial configuration exactly when the initialized QN register source halts. The finite trace translations are primitive recursive, with correctness formalizable in PA.*

Proof. The exact seed establishes the source-frame relation. Each source step has a finite, positive-duration physical segment to the prescribed next frame, or the terminal state for a source halt. The generic unary and list lemmas are proved by induction on their coded lengths and lists; the finite symbolic path certificates supply the remaining compositions. Concatenation simulates every finite source trace.

For reflection, suppose the physical computation first halts at time T. If the source has not halted within its first $T + 1$ steps, concatenating those positive nonhalting segments after the startup yields a deterministic physical trace past time T, a contradiction. This is a bounded finite argument and does not infer halting equivalence from a simulation in only one direction. □

Theorem 6.2 (Ordinary mathematical PA equivalence)**.** *Using the supplied separate-head numerical-calculus/PA proof translations,*

$$\mathrm{PA} \vdash (\neg\, \mathrm{Halt}(\mathsf{PA194QN}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})).$$

Proof. Compose the finite physical/source equivalence with the source/refutation equivalence from Section 5, then negate the existential refutation predicate. The logical Q transport and command N transport have the compatible interface established above. □

This is an ordinary mathematical proof with its inherited dependencies identified. The finite and universal symbolic executables bind the argument to this literal table. They are not newly compiled Lean/Rocq proofs, and the stronger internal iPRA project remains separate.

7 Attempts below 194 and their exact limits

No bounded search in this work yielded a verified sub-194 machine. The accepted result does not depend on running heuristic discovery again.

1. **All eligible reader folds on the QN parent.** Among its 201 controls, 130 have the required strict incoming-bit context. The inherited three-step helper template yields seven opportunities and maximum matching seven, exactly the seven used. There is no additional cross-fold with a fixed control under this template.
2. **Contraction searches on PA194HS.** A radius-six conservative window abstraction contains 269,607 configurations. The all-pairs search examines 28,755,456 trial row actions; none is accepted, and no pair is skipped. A radius-nine search restricted to pairs involving native root guard 68 has 10,719,423 abstract configurations and 297,984 trial actions, again with zero accepted and zero skipped. Both use a 64-step local horizon and require simulations to stay in their observed window.
3. **Two new logical-emitter screens.** Replacing the equality-witness head $\exists_D E(H, D)$ by $\exists_D E(A, D)$ preserves its universally instantiated axiom family but changes the emitted correlation. A bounded adaptive search ends at a projected 195 states after 2,613,698 mutations. Combining Q's exchanged sixth head with $\Gamma(D)$ and the documented known-zero conventions ends at projected 195 after 2,475,004 mutations. Neither numerical score is promoted to a consistency-machine theorem.
4. **Simpler MP outputs.** A direct transfer yielding $(b, w, 0, 0)$ is physically realized by adding a reader for $(\text{dec } D, \text{inc } T; \text{continue})$. It reaches 195 states, not 193 or 194. A cleaner command contract alone is therefore insufficient to improve the physical count.

These are finite negative results for particular rewrite, layout, and contraction classes. They establish no global lower bound. Prior rejected scanner deletions are also not general impossibility results: they fail specific source-frame obligations, sometimes after a local quotient has already passed. The report does not treat agreement on a finite blank run as evidence of nonhalting or correctness of an optimization.

7.1 Ideas from the ZFC and concise iPRA programs

The ZFC work contributes the useful no-push/explicit-reserved-zero proof compiler and the practice of stating command contracts over arbitrary carried stores. Those ideas were already imported into PA194N, so their reuse is credited rather than presented as a new transfer. Contextual decoder folding and known-zero source specializations likewise have an established history in these programs.

The concise iPRA program suggests treating rule selection and proof representation as algorithm-design choices instead of optimizing formula notation in isolation. However, replacing the PA refutation enumerator by an iPRA refutation enumerator does not preserve the requested target: it changes the consistency statement. A useful transfer must retain a PA-verifiable refutation translation and improve the compiled binary machine, not only shorten a high-level proof checker.

8 Further research questions

1. Can the root guard be specialized using its reachable input invariant without adding a reader state? In the symmetric controller one operand is always one. Generic pair-contraction failures do not

exclude a change to the source and its frame relation together.
2. Can a shorter origin scanner be obtained by jointly redesigning the source tree, temporary markers, and initialization? Bounding only the zero runs in clean source addresses misses transient-marker hazards.
3. Can a redesigned MP result convention and an AX decoder share the extra transfer reader? The 195-state direct-transfer experiment gives a concrete target whose interface is simpler than its physical allocation.
4. Can changes in different axiom heads be optimized jointly with multi-state contextual quotients? The Q+$\Gamma(D)$ search reaches a 195-state score but does not exclude a better association or a broader folding template.
5. Can the new proof-to-command compiler and inherited syntactic PA-calibration maps be expressed in the repository's explicit arithmetic coding framework, closing the current machine-specific formalization gap? This would improve assurance without confusing a shorter proof script with a smaller Turing machine.

9 Reproduction, dependencies, and provenance

The accepted reconstruction uses Python 3.10+ and its standard library. Keep assertions enabled; do not use `-O`. From the extracted packet:

```
python3 -B reproduce.py
```

The default command works in a fresh temporary copy, rebuilds the table, checks the command and exchange identities, independently resolves the source, runs the literal-tape symbolic verifier, compares the regenerated machine to the delivered bytes, and checks graph novelty. It needs no network, external repository checkout, solver, or proof assistant. Optional `-quick` skips the two full physical symbolic passes and clearly reports that reduced scope. Optional C++17 searches live under `research/`; their outcomes are not prerequisites for the theorem.

To compile this report:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The package retains the supplied licenses and source attribution. The new construction and audits build on James Chang's imported PA196 source and runtime, the ProveIt research compilers and logical calibrations, and the independent literal symbolic engines identified in their notices. No external repository changes are part of this delivery. In the references below, `PA/` abbreviates `Computability/BusyBeaver/Research/PA/` at the pinned snapshot.

References

[1] ProveIt, pinned repository `3cc7fbf75ba111a0da696ef2f6001dd528891bd5`. Primary paths: PA research, incoming deliveries, ZFC research, and ConciseIPRA.

[2] `PA/docs/pa194N_conpa_2026-11-13/`. Full no-push command proof, constructor, and physical certificates. The relevant mathematical text is retained in `references/logical/pa194n_command_proof.tex`.

[3] PA/docs/pa194q_conpa_2026-11-13/. Premise-exchange emitter, exact grammar/folds, logical transport, and 61-line exchange certificate. The calibration is retained in references/logical/pa194q_calibration.tex.

[4] PA/docs/pa194_conpa_2026-11-13/report/ and its reference/ directory. Numerical-calculus calibration and the separate diagonal-head transformation; copies are identified in provenance.

[5] docs/incoming/pa194hs_conpa_2026-11-13.zip. Self-contained compiler and independent literal-tape verifier used as the frozen runtime inputs of this construction.

[6] PA/jamesyc-upstream/pa196/PA196/Logic/ImplicationBasis.lean and PA/jamesyc-upstream/pa196/M6/Logic/RestrictedImplication.lean. Derived implication rules, including exchange, from the unchanged restricted basis.

A Complete dense transition table

An entry wDq' means write w, move in direction D, and enter q'. State 194 is terminal. The two groups of columns simply place two states on each printed line. The file `machine/machine.tm` contains the same 388 transitions in five-column form.

q	Read 0	Read 1	q	Read 0	Read 1
0	1R1	1R2	97	0R48	1R61
1	1R3	1R1	98	0R87	1L115
2	0R0	1R2	99	0R104	0L14
3	0R4	0R1	100	0R36	1R116
4	1R5	1L6	101	0R115	1R78
5	1L7	1L8	102	0L46	0R113
6	0L7	1L6	103	0L45	1R117
7	0L9	1L6	104	0L94	1R118
8	0L10	0L11	105	0R119	0L14
9	1R12	0L9	106	0R71	1R106
10	0L9	0L10	107	0R120	1R30
11	1L8	1L11	108	0L46	0R106
12	0R12	0R13	109	0R70	0R95
13	0L14	0R13	110	0R59	0R95
14	1L15	0L15	111	0L65	1R121
15	0L15	1L16	112	0R122	1R123
16	0L17	1L16	113	0R14	1R113
17	0L18	1L16	114	0R68	1R114
18	0L19	1L16	115	0R124	1L67
19	0L20	1L16	116	0R125	1R126
20	0L21	1L16	117	0R120	0L14
21	0L22	1L16	118	0R94	1L98
22	0L23	1L16	119	0R127	1R127
23	0R23	1R24	120	0R70	0R68
24	0R25	1R26	121	0R128	1R129
25	0R27	1R28	122	0R32	1R36
26	0R29	1R30	123	0R31	1R101
27	0R31	1R194	124	0R59	0R114
28	0L32	1R33	125	0R102	1R117
29	0R34	1R35	126	0R82	1R81
30	0L36	1L37	127	0R120	1R120
31	0R38	1R39	128	0R130	1R131
32	0R40	1R41	129	0R132	1R133
33	0R42	1R43	130	0R134	1R135
34	0R44	1R45	131	0R136	1R137
35	0R46	1R47	132	0R138	1R91
36	0R48	1R49	133	0R139	0L10
37	0R50	1R51	134	0R100	1R67
38	0R52	1R42	135	0R140	1R127
39	0R53	1R54	136	0R141	1R142
40	0R55	1R56	137	0R143	1R144
41	0R57	1R58	138	0R145	1R41
42	0R59	0R60	139	0R146	1R147
43	0R61	0L14	140	0L148	1L143
44	0R62	1R63	141	0R28	1R149
45	0R55	1R64	142	0R135	1R150
46	0L65	1R66	143	0R151	1R152
47	0R36	1R67	144	0R153	1R154

q	Read 0	Read 1	q	Read 0	Read 1
48	0R59	0R68	145	0L130	1R120
49	0R69	0L14	146	0R31	1R53
50	0R70	0R71	147	0R155	1R146
51	0R72	0L49	148	0R156	1R157
52	0R73	1R54	149	0R158	1R159
53	0L74	0L15	150	0L159	1R160
54	0R75	0L10	151	0R41	1L161
55	0R76	1R77	152	0R162	1R163
56	0R78	1R79	153	0R164	1R164
57	0R80	1R81	154	0R165	1R166
58	0R82	1R83	155	0R96	1R115
59	0L84	0R1	156	0R79	1R165
60	0R85	1R60	157	0R167	1R165
61	0L76	1L86	158	0R168	1R169
62	0R87	0L7	159	0R170	1R157
63	0R88	1R89	160	0R135	1R130
64	0R42	1R90	161	0R138	1R135
65	0R91	1R49	162	0R171	1R172
66	0R92	0L75	163	0R173	1R172
67	0R86	1R33	164	0L174	1R175
68	0R93	1R68	165	0L162	1L176
69	0L94	1L33	166	0R161	1R176
70	0L84	0R2	167	0R100	1R177
71	0R95	1R71	168	0R134	1R178
72	0R75	1R96	169	0R150	1R163
73	0R59	0R85	170	0R179	1R180
74	0R86	1R97	171	0R167	1R140
75	0R42	0L13	172	0R134	1R145
76	0R87	1R37	173	0R181	1R41
77	0R30	1R98	174	0R175	1R182
78	0R42	1R99	175	0R167	1R145
79	0R100	1R101	176	0R183	1R173
80	0R102	1L103	177	0R91	1R64
81	0R30	1R80	178	0R184	1R165
82	0R97	1R104	179	0R134	1R181
83	0R30	1R105	180	0R185	1R185
84	1R84	1L15	181	0R167	1R151
85	0R106	1R85	182	0R151	1R130
86	0R73	1R107	183	0R164	1R186
87	0L46	0R93	184	0R140	1R187
88	0R108	1R98	185	0R188	1L189
89	0R108	1R69	186	0R190	1R41
90	0R109	0L14	187	0R32	1R189
91	0R110	1L74	188	0R120	1R191
92	0R111	1R112	189	0R148	1R192
93	0R113	1R93	190	0R41	1R138
94	0R50	1R103	191	0R174	1R120
95	0R114	1R95	192	0R182	1R193
96	0R30	1R49	193	0R161	1R138

B All 34 prior 194-state graph classes

The descriptions classify supplied evidence. "Proof/physical" means a supplied ordinary PA argument and physical evidence, not a new formal kernel theorem checked in this work. Exact locators, graph identifiers, hashes, and retained representative tables are in `comparison/`.

Design	Evidence	Material change
Cached I-tree zero-jump seed	Auxiliary	Alternate decoder plus cached controller and entry zero jump; search basis.
CS both-zero6	Auxiliary	Known-zero X decrement at both source26 and final emitter exit; research composition.
CSZ-A3	Auxiliary	CSZ plus comparator-return register3 and final-emitter register3 decrements.
CSZ-A5	Auxiliary	CSZ plus comparator-return register3 and final-emitter register5 decrements.
CSZ-A6	Auxiliary	CSZ plus comparator-return register3 and final-emitter register6 decrements.
Gamma(D)-CSZA	Auxiliary	Supplementary Gamma(D) emitter plus known-zero changes.
M4R	Proof/physical	Reassociated coefficient4 constructor decoder; original resolved source graph.
M5	Proof/physical	Formula constructor coefficient5; explicit variable and proof recoding.
M6	Proof/physical	Formula constructor coefficient6; explicit variable and proof recoding.
MPH	Auxiliary	MP result cached in H while retaining original Shift; physical search input.
PA194	Proof/physical	Separate diagonal head, 67-reader decoder with seven folds, original controller.
PA194-68/8	Proof/physical	Same separate-head source, reassociated 68-reader decoder with eight folds.
PA194-AD	Proof/physical	PA194 with physical A/D register exchange.
PA194A	Proof/physical	CS entry jump at source26 becomes decrement of known-zero register A.
PA194altC	Auxiliary	Cached controller on alternate 68-reader decoder.
PA194C	Proof/physical	MP result remains cached in H; Shift tests H=1; original ten-entry logical interface recompiled.
PA194CS	Proof/physical	Symmetric Shift guard; cached MP controller; contextual controller fold.
PA194CSZ = PA194X	Proof/physical	CS entry jump at source26 becomes decrement X; two headline names denote identical dense bytes.
PA194CZ	Auxiliary	Cached controller; entry known-zero decrement of register 6.
PA194D	Proof/physical	Separate diagonal axiom uses Gamma(D), new proof selector, 69 readers and nine folds.
PA194F	Proof/physical	CS final emitter jump becomes decrement of known-zero register X.
PA194H	Proof/physical	No-push MP returns result in visible top; Shift tests cached H; original comparison implementation.
PA194HS = PA194NS	Proof/physical	No-push MP returns result in visible top; Shift tests cached H with symmetric comparison. Two deliveries have identical dense bytes.
PA194J	Proof/physical	Ten source jumps become known-zero decrements; three physical rows change.

Design	Evidence	Material change
PA194N	Proof/physical	Successful MP drops dummy-zero push, retains original Shift; revised command proof compiler.
PA194N5	Proof/physical	Coefficient-five formula constructors plus no-push MP; original Shift.
PA194N6	Proof/physical	Coefficient-six formula constructors plus no-push MP; original Shift. Two deliveries have identical dense bytes.
PA194Q	Proof/physical	Quantifier-distribution curried premises exchanged; K/S proof translation.
PA194R	Proof/physical	Reassociated decoder of PA194F; 1111 source paths changed, same resolved source graph.
PA194ZR	Proof/physical	Eight comparator returns and final emitter return become known-zero register3 decrements.
Rejected CS node1-X	Rejected	Native103/read1 becomes dec6; exact blank run halts at1361.
Rejected I-tree node1-X	Rejected	Known-zero source rewrite fails blank bootstrap; exact halt at1361.
Rejected seven-zero s-canner	Rejected	Decoder local folds pass, but source-address scanner violates runtime contract.
Rejected ZR node1-X	Rejected	Register6 comparator-return variant; exact blank run halts at1361.

C All 51 delivered research packets

This ledger records distinct deliveries, including repeated constructions. The full CSV/JSON versions include original paths and pinned source URLs.

Delivery	Contribution and status
pa_unary_power_cover	Unary power covering at PA strength (report of 27 October 2026). The U208 packets carry it as a baseline.
U208_ConPA_Conditional_Context_2026-10-29	U208 by a conditional tape-context reduction. States a 208-state candidate machine/u208.tm.
U208_ConPA_Entry_Context_2026-10-29	U208 by eliminating a state using entry context. States a 208-state machine machine/u208.tm.
U208_ConPA_History_2026-10-29	U208 by a finite-history decoder fold. States a 208-state candidate machine/u208.tm.
U208_ConPA_2026-10-29	U208 by a three-cell fold. States a 208-state candidate machine/u208.tm.
u208_research_2026-10-29	U208 research release. States two 208-state machines, machine/u208.tm and machine/variant_plain/u208.tm.
Below_U208_Research_2026-10-29	Research toward fewer than 208 states. States that no machine with 207 states or fewer was obtained.
u208_comparison_research_2026-10-29	Comparison of the 208-state proposals and a search below 208. States that the target was not achieved.
ConPA_below208_research_2026-10-29	Comparative audit and reduction search below 208. States that it does not establish a 207-state machine.
pa_consistency_research_2026-10-29	Audit and new source reductions. States that no validated machine below 208 states was obtained.
pa208_research_2026-10-29	Comparison and research below 208 states. States that it does not establish a valid machine below 208.
pa_tm_research_2026-11-13	Comparison and reduction research. States a new 196-state variant with shorter source paths and nothing below 196.
pa_consistency_research_2026-11-13	Comparison and diagonal-vacuity research. States a shorter axiom program with a 199-state implementation and nothing below 196.
conpa_tm_comparison_and_reduction_search	Comparison and reduction search. States two experimental 196-state command variants and nothing below 196.
pa196_research_2026-11-13	PA196 comparison, source factorization and proof obligations. States a source tree of 157 internal nodes and nothing below 196.
pa_consistency_research	Research and verification packet. States that no machine below 196 states has been validated.
pa_tm_comparison_and_reductions	Comparison and new rewrites, a research snapshot.
pa_tm_extended_research_2026-11-13	Comparison and refined reduction search. States that the smallest accepted count is 196.
conpa_tm_comparison_and_search	Comparison and a search below 196. States that none was obtained; files named 195 are rejected or diagnostic candidates.
pa_consistency_reduction	Comparison and reduction experiments. States two new 196-state variants and a shorter source program with a 203-state implementation.
pa_consistency_comparison_2026_11_13	Comparison and new reduction experiments. States that nothing below 196 was obtained.
PA_consistency_TM_research_2026-11-13	Comparison and new source reductions. States that no machine below 196 is claimed.
pa_consistency_research_2026-11-13_2	Comparison and search below 196. States two new layouts with 196 and 198 states and a command-level reduction.
conpa_tm_study_2026-11-13	Comparison and reduction study. States that nothing below 196 was obtained.

Delivery	Contribution and status
pa_below196_research	Comparison and reduction experiments. States that nothing below 196 was obtained.
pa_consistency_comparison_and_new196_2026-11-13	Comparison and reduction research. States that it contains no accepted machine below 196.
pa_tm_research_2026-11-13_2	Comparison and reduction investigation. States same-size work, failed candidates and nothing below 196.
pa_consistency_cross_archive	Cross-archive constructions and reduction search. States that it does not certify a 195-state machine.
pa195_conpa_2026-11-13	PA195. States a 195-state machine pa195/pa195.tm and an article with a PA proof of the equivalence.
pa_consistency_reduction_audit_2026-11-14	Comparison and reduction audit. States nothing below 196 and no new theorem.
pa_consistency_research_2026-11-14	Comparison and NEG18. States an 18-instruction encoded negation with a compiled machine of 197 states.
PA195_ConPA_research_2026-11-14	A 195-state machine. States the primary machine_195S.tm and a second table machine_195C.tm.
pa195_followup_research	Comparison and searches below 195. States a graph-distinct 195-state machine M195S8 and nothing below 195.
pa194_conpa_2026-11-13	A 194-state machine. States the table construction/tables/-pa194.tm and a proof of the equivalence in PA.
pa194_conpa_2026-11-13_2	PA194. States a 194-state machine pa194/pa194.tm and an article with a PA proof of the equivalence.
pa195_diagonal_combined_2026-11-13	PA195DC. States a different 195-state machine.
pa194C_conpa_2026-11-13_2	PA194C. States a materially different 194-state machine, an article with a PA proof of the equivalence, and nothing below 194. Headline PA194C is repeated across five deliveries; PA194_new_designs additionally contributes PA194CS.
pa194_new_designs_2026-11-13	Two new 194-state machines, PA194C and PA194CS. States that nothing below 194 was obtained. Headline PA194C is repeated across five deliveries; PA194_new_designs additionally contributes PA194CS.
pa194c_conpa	PA194C, a cached-head 194-state machine. States that no machine below 194 was established. Headline PA194C is repeated across five deliveries; PA194_new_designs additionally contributes PA194CS.
pa194c_conpa_2026-11-13_1	PA194C. States a different 194-state machine and an article with a PA proof of the equivalence. Headline PA194C is repeated across five deliveries; PA194_new_designs additionally contributes PA194CS.
pa194c_conpa_2026-11-13	PA194C, a cached-head 194-state machine. States an article with a PA proof of the equivalence and nothing below 194. Headline PA194C is repeated across five deliveries; PA194_new_designs additionally contributes PA194CS.
pa194_radix_family_2026-11-13	New 194-state machines calibrated to Con(PA), a radix family.
pa194j_pa194d_conpa_2026-11-13	PA194J and PA194D. States two new 194-state machines.
pa194q_conpa_2026-11-13	PA194Q and PA194CSZ. States two new 194-state machines and that no machine below 194 was established.
PA194R_research_2026-11-14	PA194R. States a different 194-state machine with a new decoder layout and that no machine below 194 was verified. PA194X equals PA194CSZ byte for byte; R, F, and A are distinct.
pa194N_conpa_2026-11-13	PA194N. States a new 194-state machine, an article with a PA proof of the equivalence, and nothing below 194.

Delivery	Contribution and status
`pa194zr_conpa_2026-11-13`	PA194ZR. States a new 194-state machine machine/pa194zr.tm, a PA proof of the equivalence, and nothing below 194.
`pa194hs_conpa_2026-11-13` *Incoming ZIP*	PA194H and PA194HS; successful MP omits dummy-zero push and leaves result in visible T; Shift tests cached H. HS uses symmetric comparison. Both 194 states. PA194HS is byte-identical to PA194NS.
`pa194n5_n6_conpa_2026-11-13` *Incoming ZIP*	PA194N5 and PA194N6; coefficient-5/6 formula constructors combined with no-push MP. Two 194-state machines. Node verifier supplied. PA194N6 is byte-identical to the N6 machine in pa194n6_research_2026-11-13.
`pa194n6_research_2026-11-13` *Incoming ZIP*	PA194N6; coefficient-six formula constructors combined with no-push MP, 3776 sites and 4877 primitive outcomes, independently audited literal runtime. The headline N6 table is a duplicate of the N5/N6 submission.
`pa194ns_conpa_2026-11-13` *Incoming ZIP*	PA194NS; no-push MP and symmetric cached-head Shift, with all-store compiler and 4750 primitive outcomes. No machine below 194. PA194NS is byte-identical to PA194HS.